

EXERCISE 2

01. SELECT DISTINCT(department)
FROM students;

Output:

department
IT
HR
Finance

02. SELECT department,
AVG(Age) AS average-age
FROM students
GROUP BY department;

Output:

department	average-age
IT	20.5
HR	22
Finance	23

03. SELECT department,
COUNT(DISTINCT(student_id)) AS
student-count
FROM students
GROUP BY department
HAVING student-count > 1

Output:

department	student-count
IT	2
HR	2

04. SELECT student-id,
name,
age,
department
FROM students
WHERE age BETWEEN 21 AND 23;

Output:

student-id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

05. SELECT student-id,
name,
age,
department
FROM students
WHERE department = 'IT'
OR department = 'HR'
AND age > 21 ;

Output:

student-id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

06. SELECT department,
SUM(credits) AS total-credits
FROM courses
GROUP BY department
HAVING total-credits > 5;

Output:

department	total-credits
IT	11

07. SELECT course-id,
course-name,
department,
credits
FROM courses
WHERE NOT credits = 4;

Output:

course-id	course-name	department	credits
101	SQL basics	IT	3
104	Excel	finance	2
105	Statistics	HR	3

08. SELECT course-id,
course-name,
credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

Output:

course-id	course-name	credits
102	Python	4
103	Data Science	4
101	SQL basics	3

09. SELECT MAX (grade) AS max-grade,
 MIN (grade) AS min-grade,
 AVG (grade) AS avg-grade
 FROM enrollments;

Output:

max-grade	min-grade	avg-grade
90	78	84.6

10. SELECT course-id,
 COUNT (*) AS enrollment-count
 FROM enrollments
 GROUP BY course-id;

Output:

course-id	enrollment-count
101	1
102	1
103	1
104	1
105	1

11. SELECT department,
 SUM (salary) AS total-salary,
 SUM (bonus) AS total-bonus
 FROM salaries
 GROUP BY department;

Output:

department	total-salary	total-bonus
IT	122000	10500
HR	109000	7500
finance	70000	6000

12. SELECT department,
 AVG(salary) AS avg-salary
 FROM salaries
 GROUP BY department
 HAVING avg-salary > 55 000

Output:

department	avg-salary
IT	61 000
Finance	70 000

13. SELECT employee-id,
 name,
 salary,
 bonus,
 salary + bonus AS total-compensation
 FROM salaries
 WHERE salary + bonus > 60 000 ;

Output:

employee-id	name	salary	bonus	total-compensation
1	Tom	60 000	5000	65 000
3	Spike	70 000	6000	76 000
4	Tyke	62 000	5500	67 500

14. SELECT department,
SUM(budget) AS total-budget,
AVG(budget) AS avg-budget
FROM projects
GROUP BY department
HAVING avg-budget > 70 000 ;

Output:

department	total-budget	avg-budget
IT	270 000	135 000
Finance	80 000	80 000

15. SELECT project-id,
project-name,
department,
budget
FROM projects
WHERE budget BETWEEN 50 000 AND
120 000 AND NOT department = 'Marketing';

Output:

project-id	project-name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000